

December 2025

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

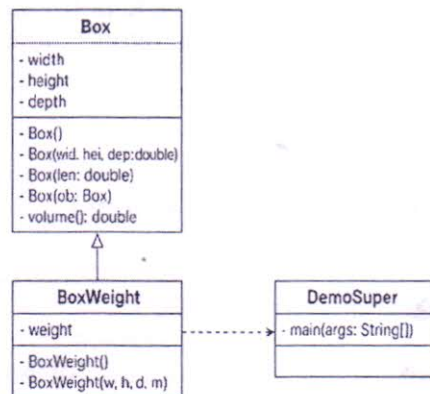
20 Marks

- 1-

10. To insert, update, or delete data, which method of Statement is used?
 A) executeQuery() B) executeUpdate()
 C) executeInsert() D) executeModify()
11. JDBC stands for:
 A) Java DataBase Compiler B) Java DataBase Connectivity
 C) Java Data Binding Connector D) Java Database Controller
12. SQL is used in JDBC for:
 A) Database connectivity only B) GUI development
 C) Querying and modifying databases D) Compiling Java code
13. A URL typically contains:
 A) Protocol, host, port, and resource path B) Only the IP address
 C) MAC address and DNS name D) Port number and DNS name only
14. The InetAddress class in Java is used to:
 A) Represent port numbers B) Represent IP addresses and hostnames
 C) Handle socket communication D) Encrypt network messages
15. A TCP socket in Java is represented by which class?
 A) Socket B) DatagramSocket
 C) ServerSocket D) InetAddress
16. To send data over a TCP socket, which stream is used?
 A) InputStream B) OutputStream
 C) FileOutputStream D) PrintWriter only
17. JSP pages are internally converted into:
 A) HTML files B) Java Servlets
 C) Class libraries D) JavaBeans
18. Which of the following is NOT a type of JSP tag?
 A) Directive tag B) Declaration tag
 C) Expression tag D) Execution tag
19. To get multiple values of a form parameter, which method is used?
 A) getParameter() B) getAttribute()
 C) getParameterValues() D) getAllParameters()
20. Which object is an implicit object in JSP used to access session data?
 A) request B) response
 C) session D) application

PART - B: DESCRIPTIVE ANSWER QUESTIONS

- | Unit – I | | Marks | BT* | CO* | PO* |
|----------|--|-------|-----|-----|-----|
| 1. | a) Explain Method Overloading and Method Overriding with suitable examples. | 10 | L*2 | 1 | 1,2 |
| | b) Write a small example showing how to create and use a JavaBean in a Java program. | 6 | L3 | 1 | 1,2 |
| 2. | a) | | | | |



Write a Java program implementing the above class diagram. Your program should:

- Define the Box class with all the given constructors and volume() method.
 - Define the BoxWeight class extending Box, with additional constructors using super().
 - Define a DemoSuper class with the main() method to create different objects and display their volume and weight.
- 10 L3 1 1,2
- b) Write a Java program to demonstrate the use of all four StringBuffer constructors. Display the length and capacity for each object created.
- 6 L3 1 1,2
3. a) What are random access files? Explain with a Java program how to write and read fixed-length records using RandomAccessFile.
- 10 L3 2 1,2
- b) Write a Java program that:
- Prompts the user to enter a file name.
 - Accepts 5 student marks from the user.
 - Validates that each mark is between 0 and 100; if invalid, re-enter the mark.
 - Writes only the valid marks into the specified file.
 - Displays a confirmation message once writing is complete.
- 6 L3 2 1,2

Unit – II

4. a) What are transactions in JDBC? Explain the concepts of commit, rollback, and savepoint with examples.
- 6 L2 3 1,2
- b) How can you modify the contents of a database (insert, update, delete) using JDBC? Write a sample program to update the salary of an employee in an Employees table.
- 10 L3 3 1,2
5. a) Explain with an example how client-server communication takes place using sockets.
- 6 L2 3 1,2
- b) Explain how Java methods can be used to modify database contents (through PreparedStatement and CallableStatement). Give an example of calling a stored procedure from Java.
- 10 L3 3 1,2
6. a) Write a note on the methods of InetAddress class such as getLocalHost(), getByName(), and getAllByName().
- 6 L2 4 2
- b) Compare TCP and UDP protocols. Write a Java program that demonstrates communication between two systems using UDP sockets.
- 10 L3 4 2

Unit – III

7. a) Write and explain a simple Java Servlet program that displays "Hello, World!" on a web page.
- 8 L2 5 2
- b) What are HTTP Request Headers? List any five important request headers and explain their significance in client-server communication.
- 8 L2 5 2
8. a) Explain the process of form handling in JSP. Differentiate between doGet() and doPost() methods in form submission.
- 8 L2 5 2
- b) Write a JSP page that accepts a user's name and email through an HTML form and displays them on another JSP page.
- 8 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
